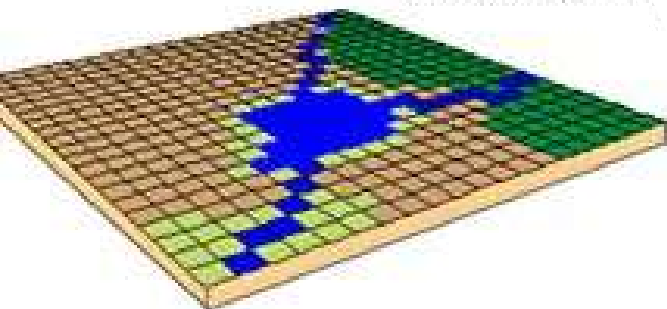
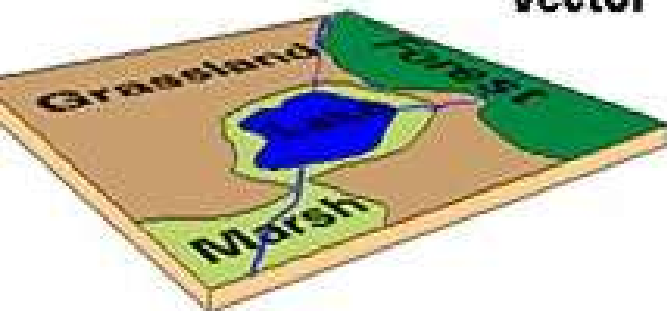


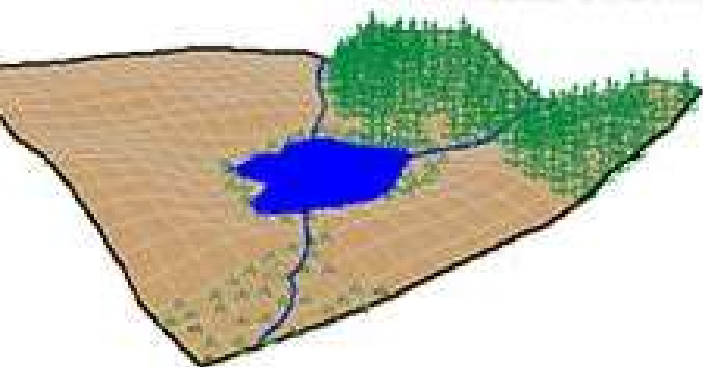
Raster / Index



Vector



Real World



Spatial Data in R

Nicole Barbour, PhD

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What is "spatial data"?

What is "spatial data"?

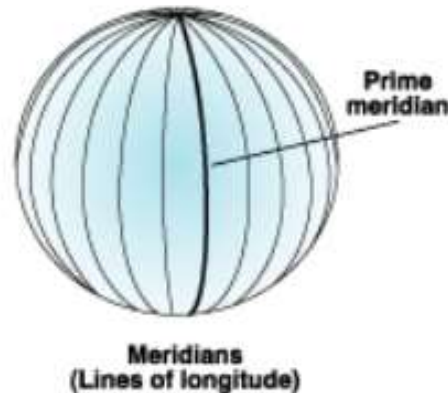
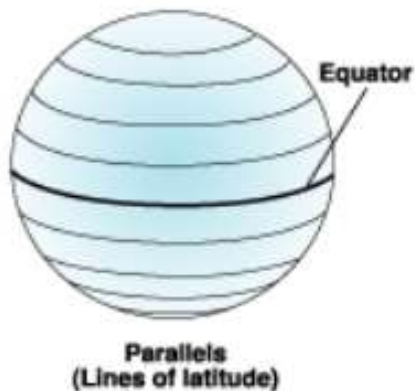
... Data with locational information!

Has:

- **spatial** attributes (e.g. geographic location)
- **non-spatial** attributes (e.g. id name, quantitative data)

Geographic Coordinates

- Locations on Earth's surface have *geographic coordinates* or positions on the latitude (x) and longitude (y) graticule



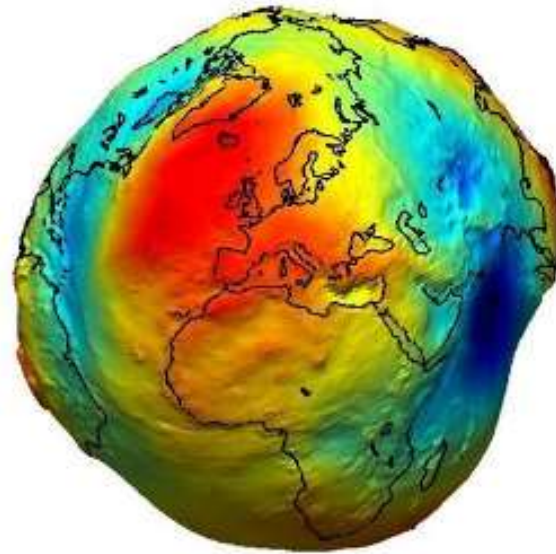
Geographic Coordinates

... so, if someone gives you a set of coordinates (latitude, longitude) can you tell me *exactly* where on Earth it is?

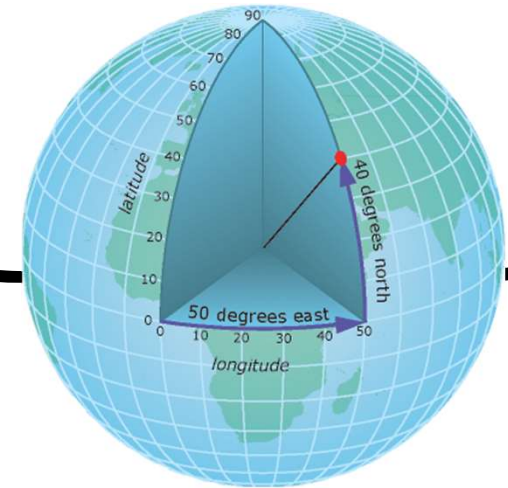
Geographic Coordinates

Probably not...

Earth is an *imperfect*,
spinning sphere!



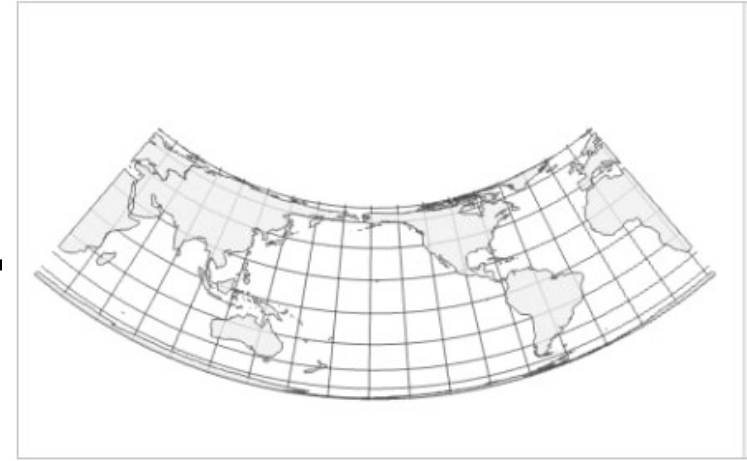
Geographic Coordinates



Geographic Coordinate Systems (GCS) represent different *models* of Earth's surface and relate graticule (lat, long) values to real locations on Earth

E.g., WGS 1984 (World Geodetic system 1984) is a coordinate system great for mapping global data!

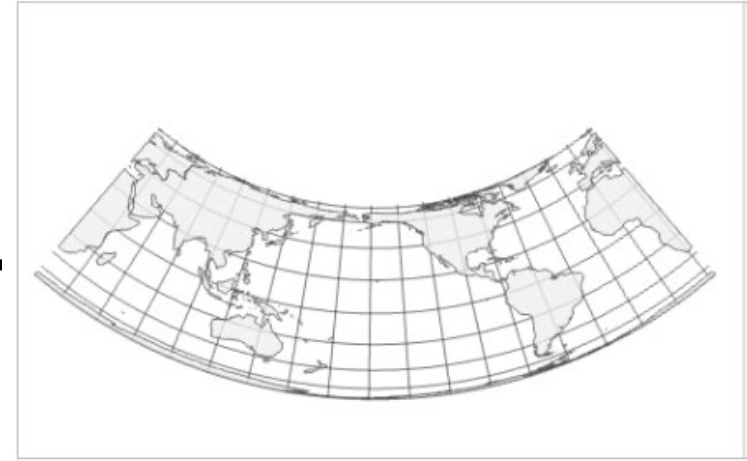
Geographic Coordinates



Projected Coordinated Systems (PCS) allow you to draw a section of a round Earth on a flat piece of paper!

PCS's are basically just flattened GCS's... so you need to find the right projection for the location of your data!

Geographic Coordinates



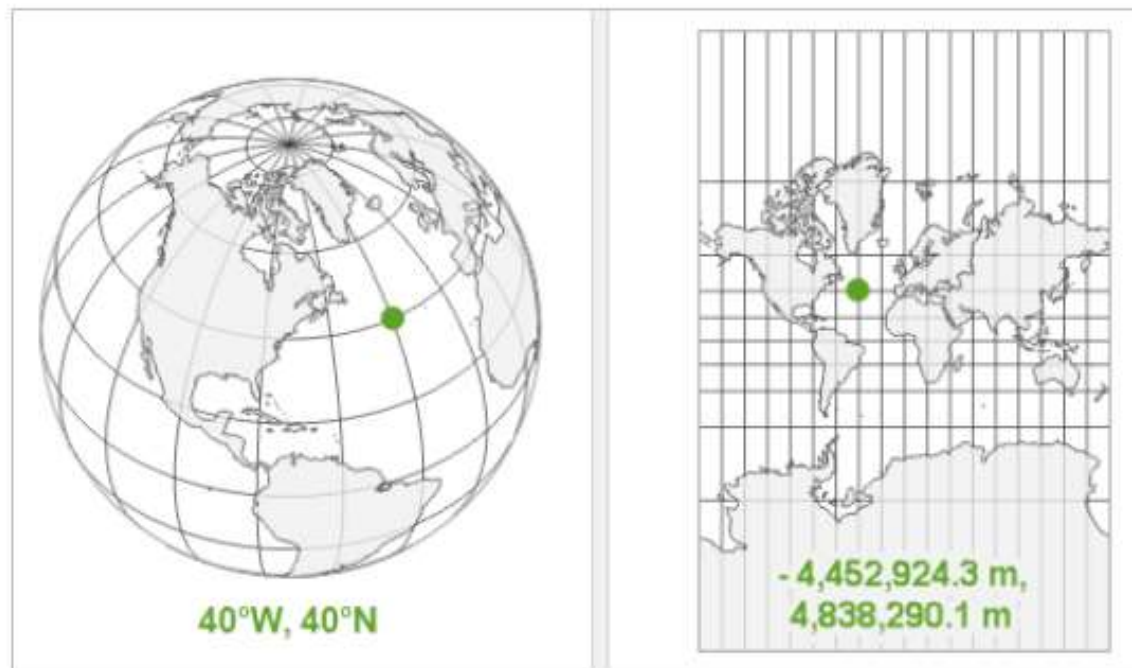
Projected Coordinated Systems (PCS) allow you to draw a section of a round Earth on a flat piece of paper!

E.g., the Universal Transverse Mercator (UTM) system is a projection that divides the globe into zones

Geographic Coordinates

GCS:

- Units are angular degrees
- Latitude, Longitude



PCS:

- Units are linear measures
- E.g., meters

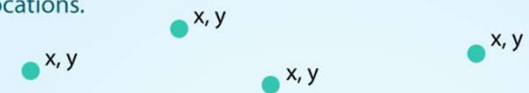
What is "spatial data"?

Also can have:

- **size** (*length, area, perimeter*)
- **shape** (*point, line, polygon*)
- **spatial relationships**

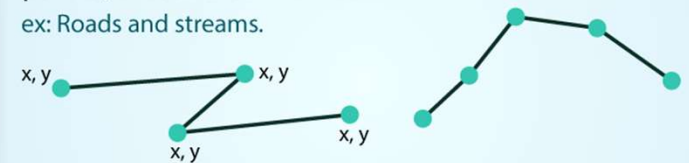
POINTS: Individual x, y locations.

ex: Center point of plot locations, tower locations, sampling locations.



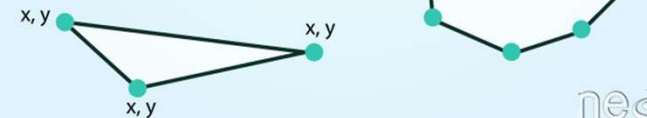
LINES: Composed of many (at least 2) vertices, or points, that are connected.

ex: Roads and streams.



POLYGONS: 3 or more vertices that are connected and **closed**.

ex: Building boundaries and lakes.



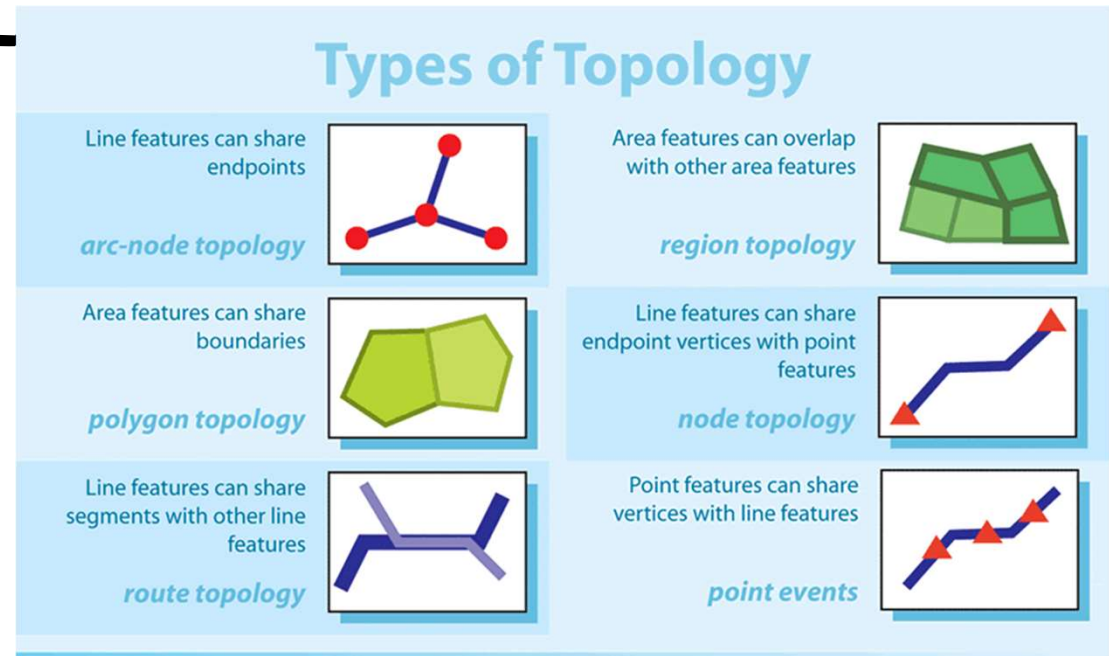
neon

What is "spatial data"?

Also can have:

- size
- shape
- **spatial relationships:**

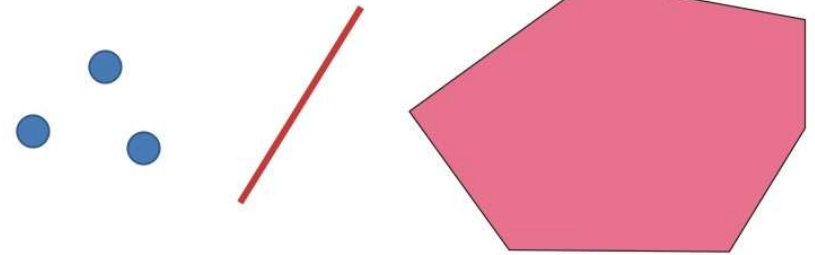
adjacency, connectivity, overlap, proximity...



Vector vs. Raster Data

- Vector data: *graphically represent “real-world” features*

... points, lines, & polygons



Points

Line

Polygon

* *Often stored as shapefiles*

Vector vs. Raster Data

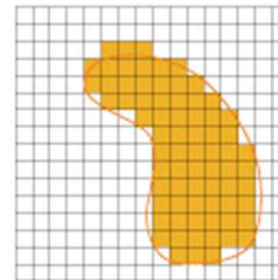
- Raster data: *represent data as a grid of pixels*

... pixels have value & resolution

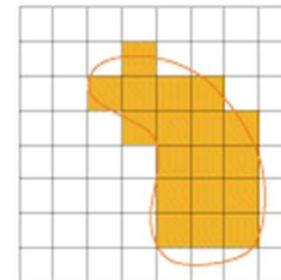


OID	VALUE	COUNT	TYPE	AREA	CODE
0	1	9	Forest land	8100	FL010
1	2	5	Wetland	4500	WL001
2	3	9	Crop land	8100	CL301
3	4	11	Urban	9900	UL040

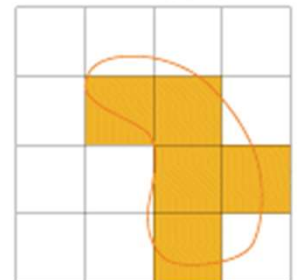
73 m²
1 m cell
16 x 16 cells



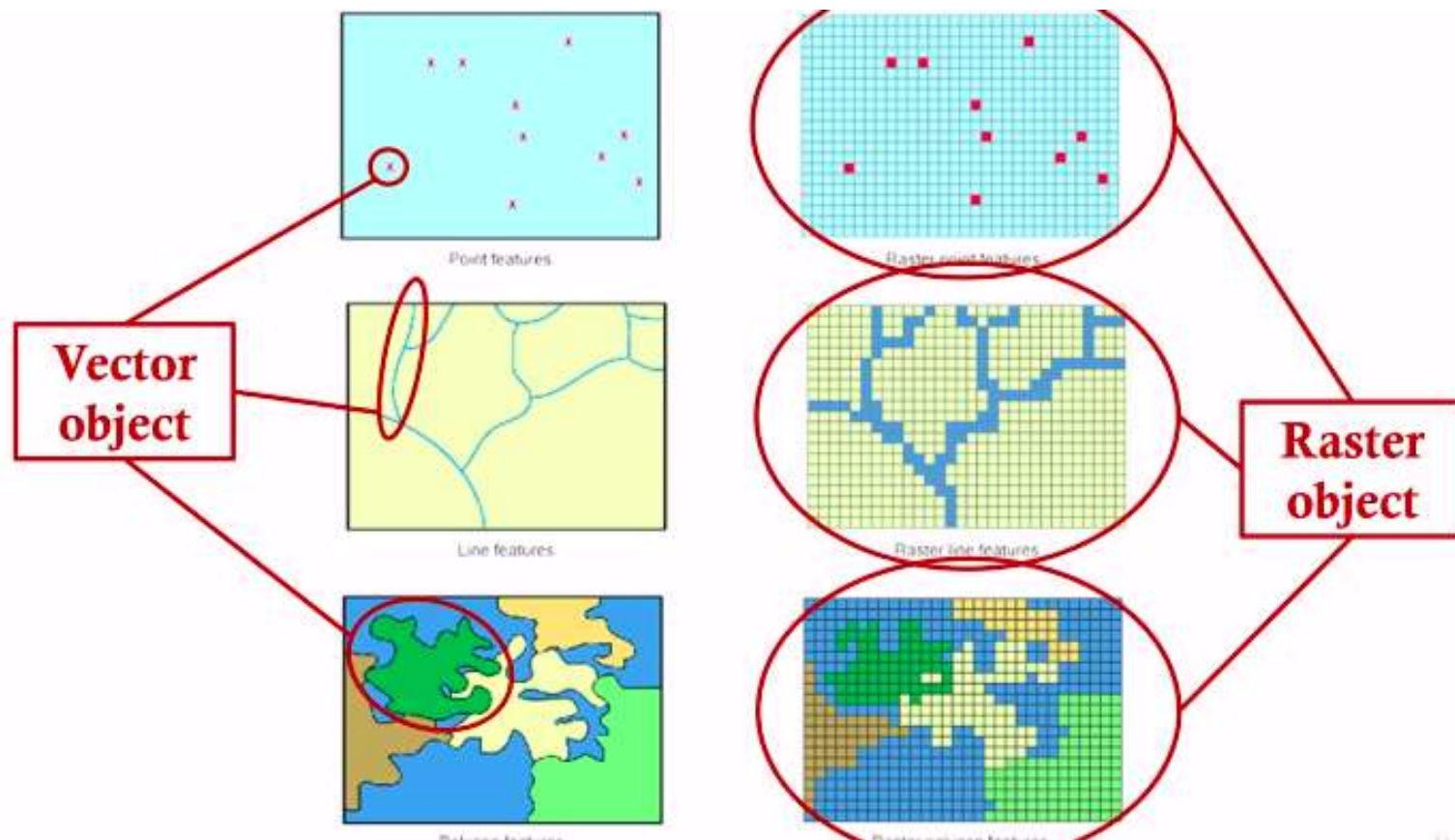
72 m²
2 m cell
8 x 8 cells



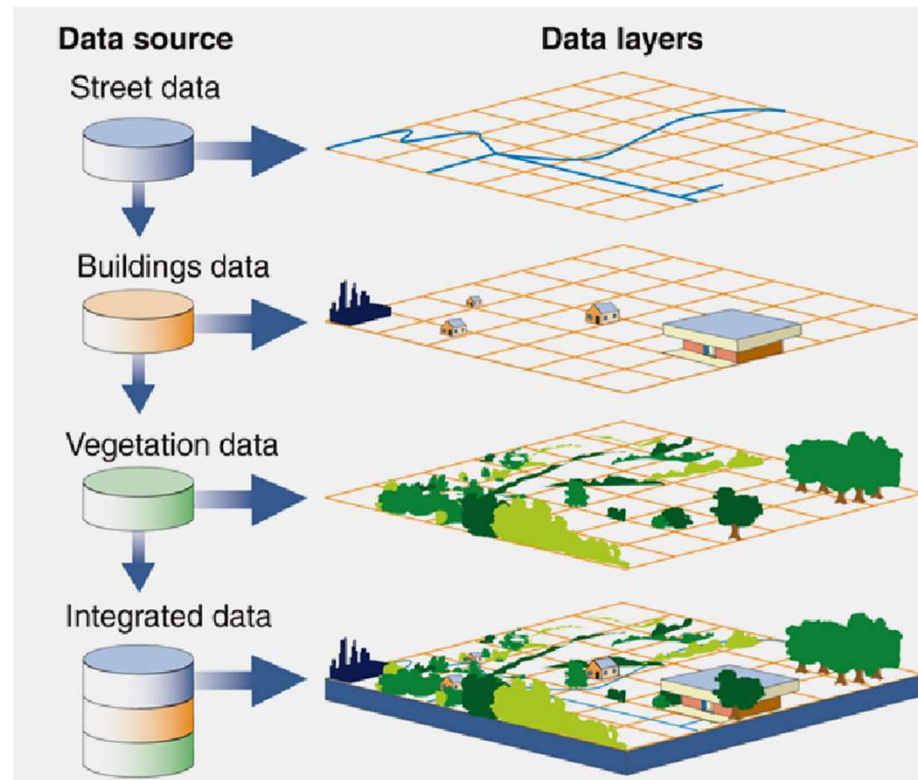
80 m²
4 m cell
4 x 4 cells



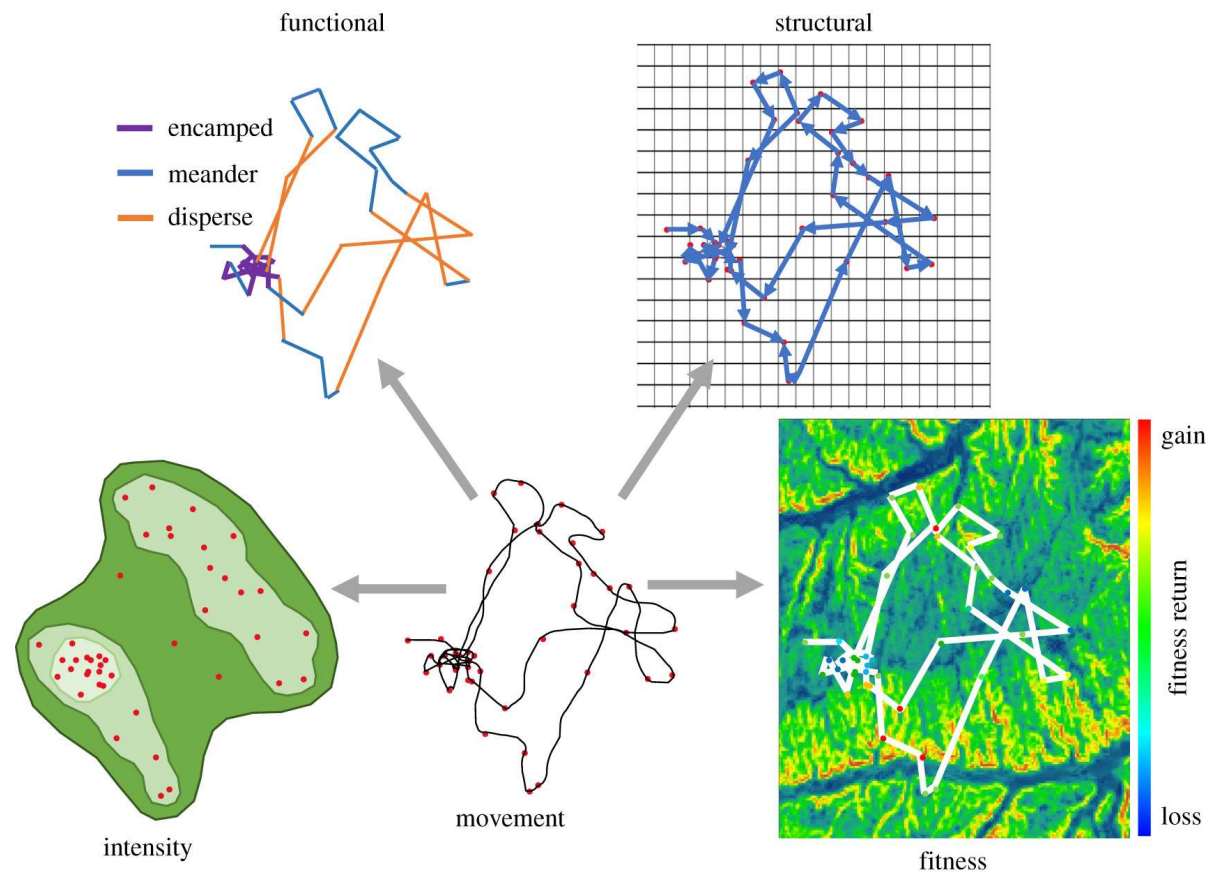
Vector vs. Raster Data



Why do we need to know spatial relationships?



Why do we need to know spatial relationships?



Tips and Tricks: Spatial Data in R

- You need to define the **coordinate system** and/or **projection** of your data before mapping/doing spatial stuff with it
 - Make sure it matches what the metadata says! Otherwise you get this...



Tips and Tricks: Spatial Data in R

- To do *quantitative operations* with spatial data, you usually want it in a PCS with measurable units
- If you want to convert from a GCS to PCS or vice versa, you need to **transform** your spatial data
 - There are fancy functions in R for this, you just need to know the code for what you want to transform it to!
 - Make sure all your data layers you are mapping or performing spatial operations with are the SAME CRS/PRS

Tips and Tricks: Spatial Data in R

- When working with *spatial relationships*, it's important to know the **resolution** of your data and what's appropriate for your research questions
 - This can prevent issues with interpolation/extrapolation
 - This is especially true when working with different spatial data types (rasters and vectors) OR when trying to merge/stack data with different resolutions
 - You can change raster resolutions and extents also with some fancy R functions...
 - Large rasters with high resolutions can take a LONG time to process/plot! Sometimes converting your data to vectors (e.g. multipolygons) is easier...

Tips and Tricks: Spatial Data in R

- ...all of which to say: it's very important to visualize your spatial data, check your data structure after performing spatial operations, and make sure that it all MAKES SENSE
 - Ask yourself: “is **MY** turtle in the ocean?... **SHOULD** it be?”



Tips and Tricks: Spatial Data in R

- When in doubt... Google it!
- There are lots of great forums, posts, websites, books, etc about working with spatial data in R
 - Packages are constantly evolving... ones that I use regularly/currently are sf, raster, ggspatial, mapview, and geosphere



In Class Exercises in R

- Download the data and code!